EmoQ: An Emotion-Aware Desktop Companion Robot for Elders

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Project Overview

Why EmoQ?

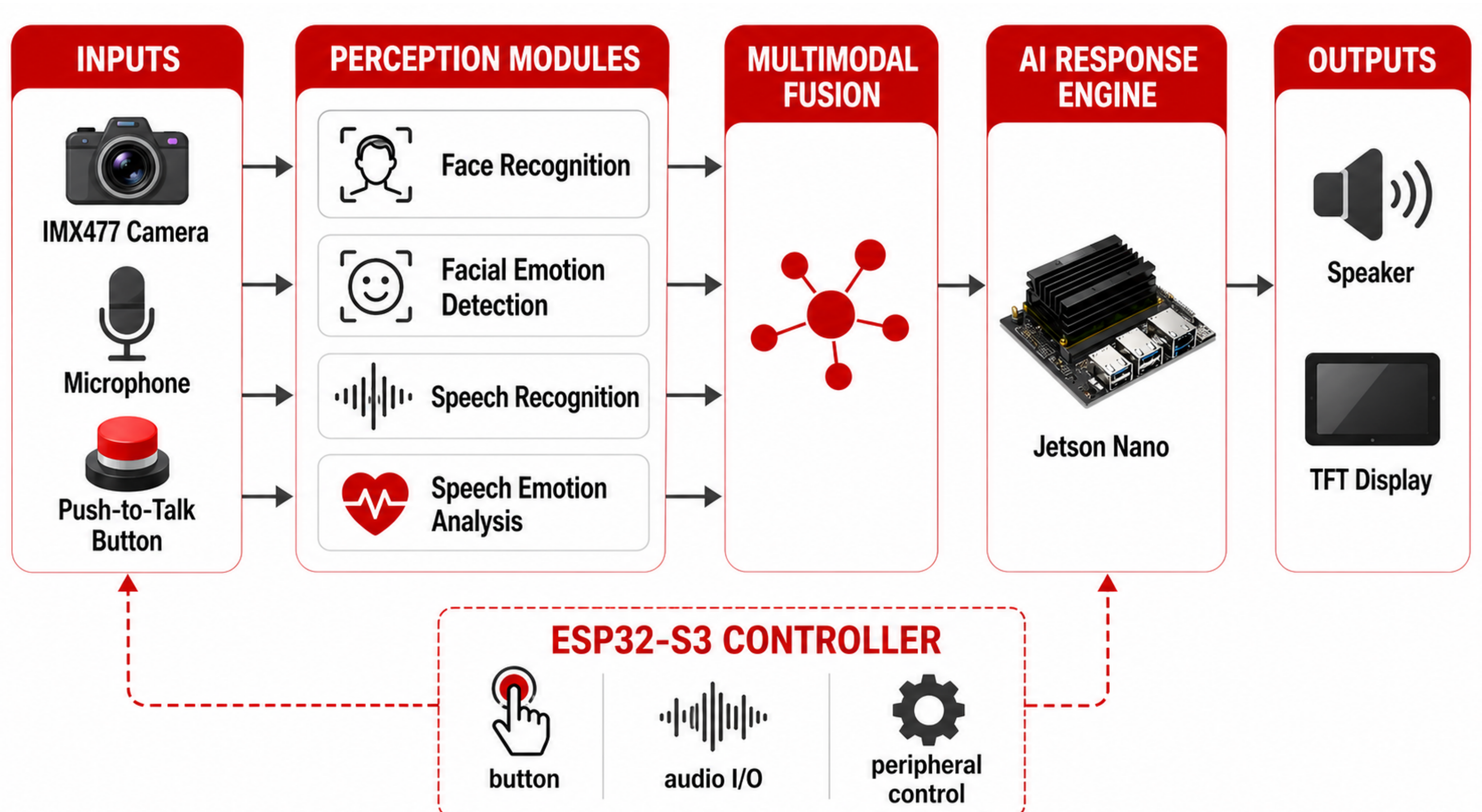
Problem: around 1 in 4 community-dwelling adults aged 65+ are socially isolated, making daily companionship an important need.

Gap: most assistive devices respond to commands, but do not understand user identity, emotional state, or interaction context.

Goal: build a desktop robot that listens, recognizes users, senses emotion, and responds with voice and visual feedback.

Listen	Understand	Respond
speech input	identity + emotion	voice + display

System Architecture



Main Contributions

- Multimodal emotion understanding through speech, face, and voice cues.
- Personalized interaction through facial recognition and registered user names.
- Robust fusion across visual emotion, speech emotion, and user identity.
- Emotionally aligned AI response generation for supportive conversation.
- Push-to-talk control for reliable, intentional, and privacy-aware speech input.
- Real-time monitoring and conversation modes for natural human–robot interaction.
- Integrated desktop robot prototype with screen, camera, microphone, and speakers.
- Elder-centered design focused on accessibility, comfort, and emotional support.

Emotion Perception: How EmoQ Understands the User

Voice Understanding

- Microphone captures the user's spoken input
- Speech recognition converts voice into text
- Speech emotion analysis estimates vocals
- Output: text + speech emotion confidence

Face Understanding

- Camera detects the user's face in real time
- Face recognition checks whether the user is registered
- Facial emotion detection estimates visual expression
- Output: identity + facial emotion confidence

Why Multimodal?

A single input can be unreliable. Speech may be noisy, and facial expression may change with lighting or camera angle. EmoQ combines both channels to make emotion perception more stable.

Voice Emotion

tone, speech pattern, intensity

+

Facial Emotion

expression, face movement, visual cues

⇓

Estimated User State

Multimodal Perception and Fusion

Inputs to Fusion

$$\mathbf{p}^{(a)} = [P^{(a)}_{happy}, P^{(a)}_{neutral}, P^{(a)}_{sad}]$$
$$\mathbf{p}^{(v)} = [P^{(v)}_{happy}, P^{(v)}_{neutral}, P^{(v)}_{sad}]$$
$$c_a,\ c_v = \text{audio / visual confidence}$$
$$id = \arg\max_i s(\mathbf{e}, \mathbf{e}_i)$$

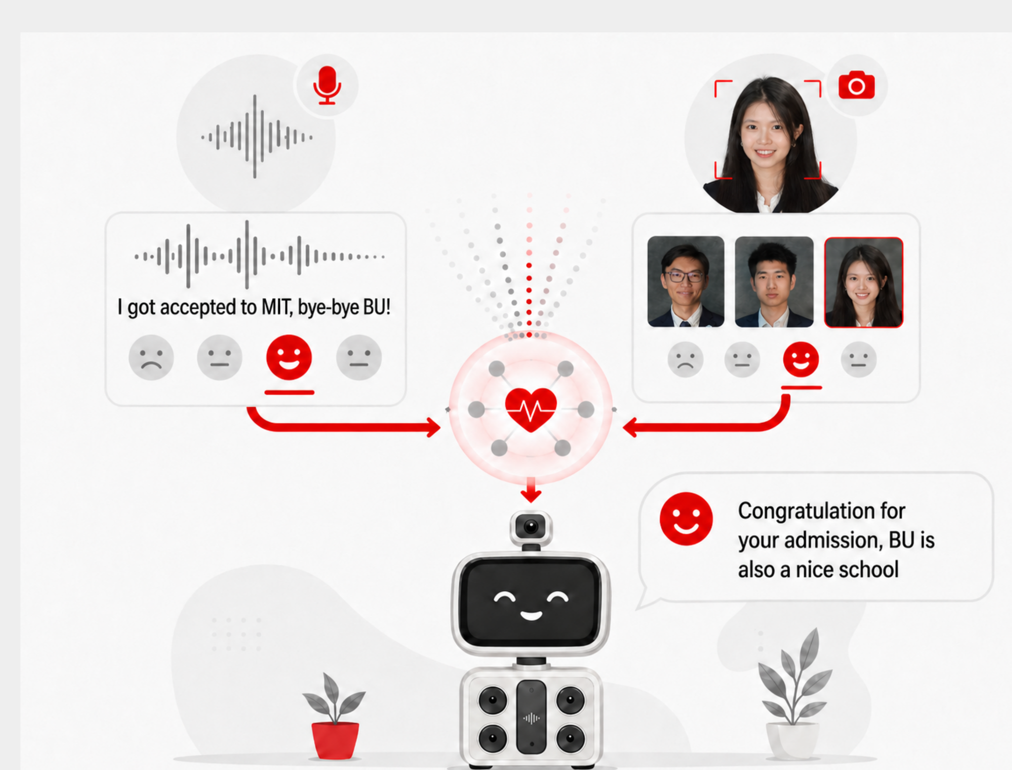
Audio and visual modules output emotion probabilities; face recognition returns the closest registered identity.

Fusion and Response

$$\lambda = \frac{c_a}{c_a + c_v}$$
$$\mathbf{p}^{(f)} = \lambda \mathbf{p}^{(a)} + (1-\lambda)\mathbf{p}^{(v)}$$
$$\hat{e} = \arg\max_k \mathbf{p}^{(f)}_k$$

$(text,\ id,\ \hat{e}) \rightarrow$ AI Response $\rightarrow$ Voice

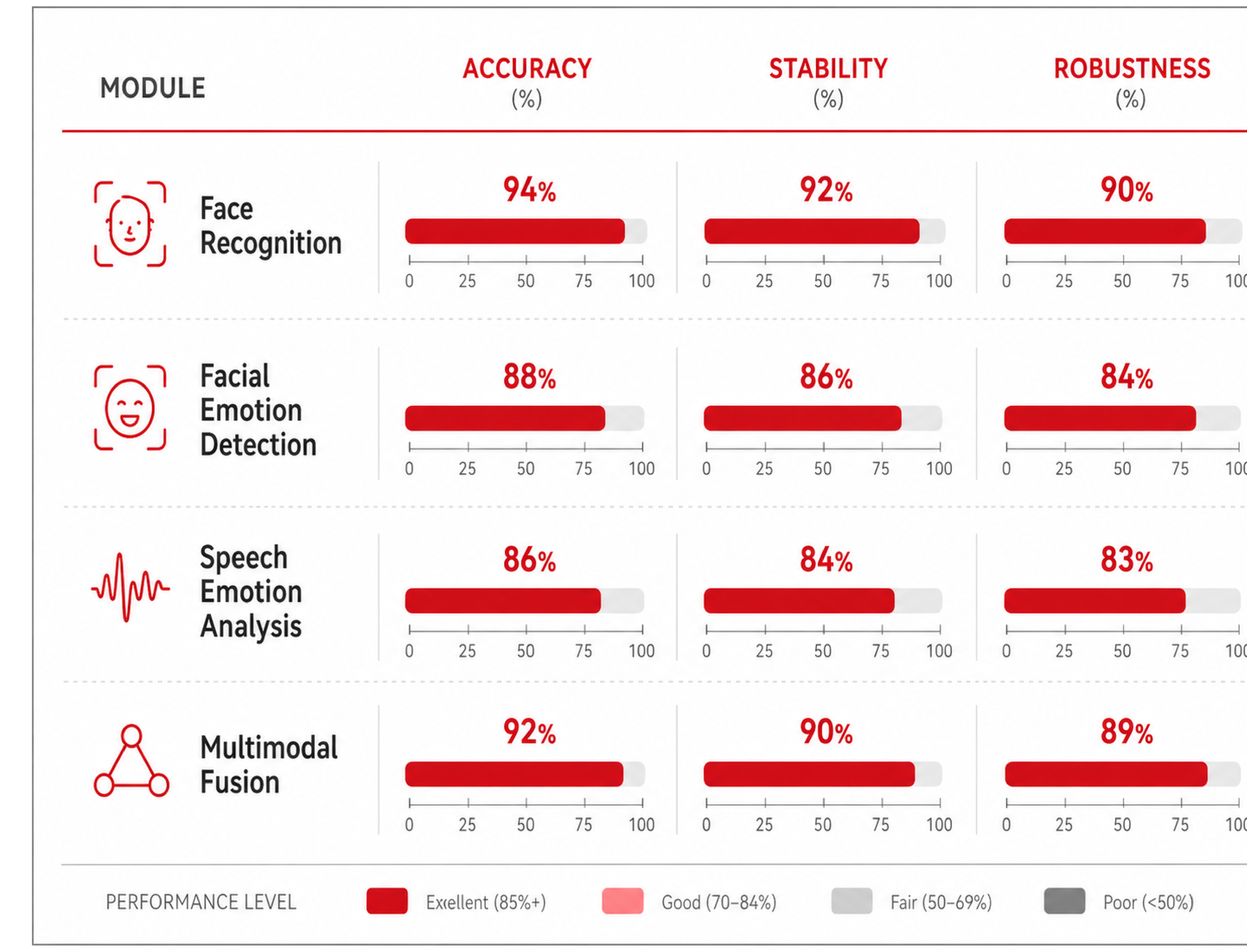
User Interaction Flow



EmoQ follows a five-step interaction loop:

1. User speaks.
2. EmoQ recognizes the user.
3. EmoQ detects emotion.
4. AI generates a response.
5. EmoQ speaks back.

Prototype Results and Design Evaluation



Limitations and Future Work

Limitations	Future Work
noise affects speech input; lighting changes affect face detection	add microphone filtering; improve camera exposure handling
face angle sensitivity; limited emotion categories	expand training data; refine emotion confidence scoring
cloud response latency; occasional AI misunderstanding	reduce response delay; add safer response filtering
visible wiring; prototype-level enclosure	improve cable management; polish product enclosure

Conclusion

EmoQ demonstrates a complete multimodal companion robot pipeline that connects speech, vision, identity recognition, emotion detection, and AI-generated response. The final prototype shows how emotional awareness and personalized interaction can make desktop assistive technology more engaging, accessible, and human-centered. Future improvements can further strengthen reliability, response speed, and long-term usability in everyday home environments.

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